Simulating steady state spectra of Acrolein while accounting Nuclear Quantum Effects

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Abstract: In the following work we discuss the molecular spectrum of Acrolein obtained using adaptive-Quantum Thermal Bath (ad-QTB) method. This method has shown in previous works to include nuclear quantum effects (NQEs) at the cost of classical molecular dynamics calculations. In this work we intend to observe, if or not, we can overcome the zero-point energy leakage problem in QTB method by strictly following Fluctuation Dissipation Theorem (FDT) to obtain a spectrum.

Introduction

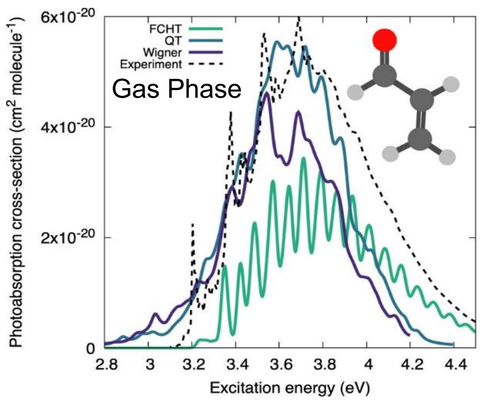
Objective:

- Simulate Nuclear Quantum effects using state-of-the art sampling techniques
- Proof of concept to capture NQE using adQTB

Why Acrolein ? Good model system with hydrogen bonding (with water)

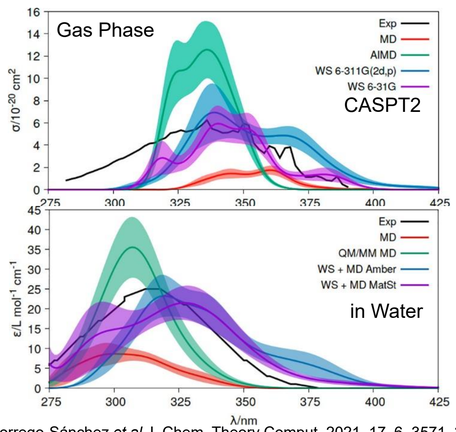
- And previous results are still not completely concluding :
  - MD results are not good
  - Wigner sampling improves the results. Is it because of nuclear quantum effects - notably Zero-Point Energy (ZPE) ?

Previous studies of gas phase and in water photoabsorption spectrum.

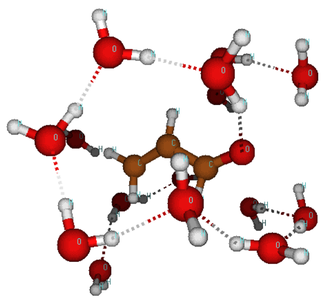


TD-DFT/TDA/PBE0/6,311G*

A. Prlj *et al* ACS Earth Space Chem. 2022, 6, 1, 207–217

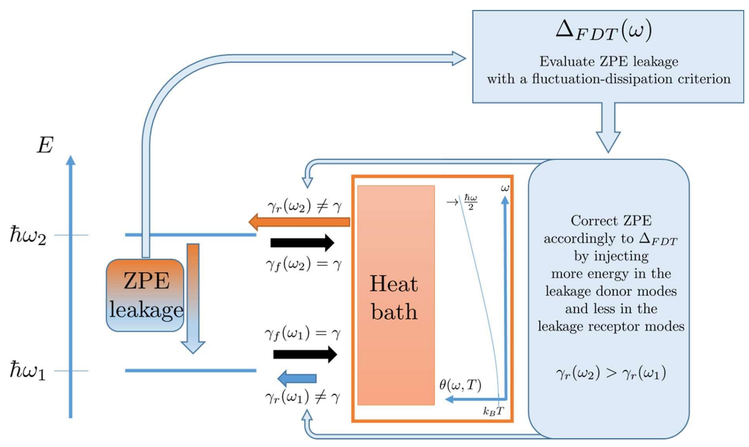


A. Borrego-Sánchez *et al* J. Chem. Theory Comput. 2021, 17, 6, 3571–3582



Method

A novel method is introduced to capture Nuclear quantum effects at cost of classical MD



Langevin (Classical) LGV Dynamics

Each nuclear degree of freedom follows an equation : $m\dot{\boldsymbol{v}} = -m\gamma\boldsymbol{v} - \boldsymbol{\nabla} V(\boldsymbol{R}) + \boldsymbol{F}(t)$

$V(\boldsymbol{R})$ → Potential energy $F(t)$ → Random Forces γ → Friction Coeff

Fluctuation Dissipation Theorem (FDT) $C_{FF}(\omega) = \langle F(t)F(t')\rangle = 2\gamma m k_B T$

R Kubo 1966 Rep. Prog. Phys. 29 255

Quantum Thermal Bath (QTB) Dynamics

Accounts for the ZPE as for quantum harmonic oscillators $C_{FF,QTB} = 2m\gamma\Theta(\omega, T)$ → Colored Noise

With Bose distribution; $\Theta(\omega, T) = \hbar\omega\left(\frac{1}{2} + \frac{1}{\left(1 - e^{\frac{\hbar\omega}{k_B T}}\right)}\right)$

BUT ZPE leakage !!

Barrat, JL., Rodney, D. *J Stat Phys* **144**, 679–689 (2011)

Dammak et al Phys. Rev. Lett. **103**, 190601 (2009)

Adaptive Quantum Thermal Bath (ad-QTB)

Find ZPE leakage by Quantum fluctuation dissipation theorem

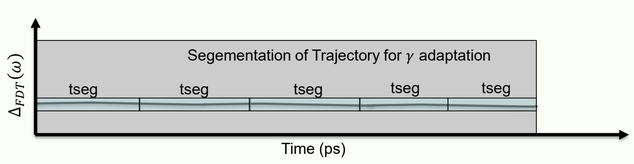
$$\Delta_{FDT}(\omega) = mC_{vv}(\omega)\gamma(\omega) - Re[C_{vF}(\omega)]$$

Reduce ZPE leakage by counterbalancing its effects in the correlation spectrum noise

$$C_{FF,adQTB} = 2m\gamma(\omega)\Theta(\omega, T)$$

With an adaptive scheme on a segment of trajectory k

$$\gamma^{k+1}(\omega) = \gamma^k(\omega) - A_\gamma \overline{\Delta}^k_{FDT}(\omega) \qquad A_\gamma \rightarrow \text{adaptation coefficient}$$



Mangaud *at al* J. Chem. Theory Comput. 2019, 15, 2863−2880

Mauger *at al* J. Phys. Chem. Lett. 2021, 12, 34, 8285–8291

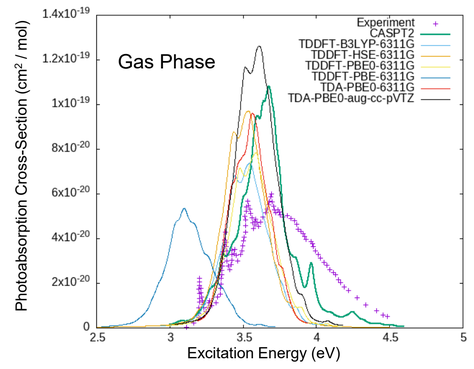
Ple *et al* J. Chem. Theory Comput. 2023, 19, 1432−1445

Implementation and challenges

adQTB is implemented in **Tinker-HP** Software.

Parametrization:

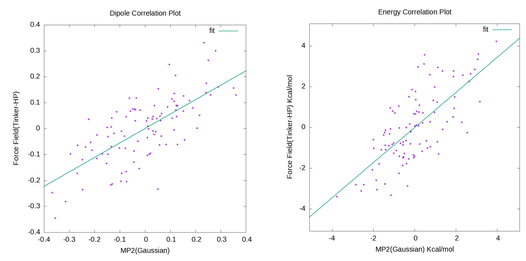
- A good force field
- Integrator and Equilibration
- Selection of friction term (all methods)
- Selection of (ad)QTB parameters
  - ❑ Friction coefficient
  - ❑ Segmentation length (tseg)
  - ❑ Adaptation Coeff A_γ (ad-QTB)
- Electronic Structure method

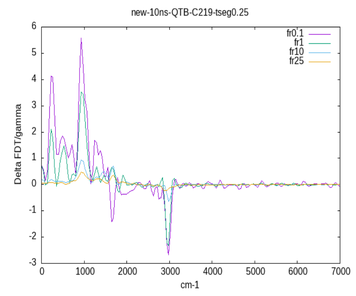


Electronic Structure Methods, TDA-DFT with different functionals compared with CASPT2

ad-QTB γ adaptation analysis shows the reduction of leakage for all the modes.
Equilibration time: 40ns
Production: 10ns

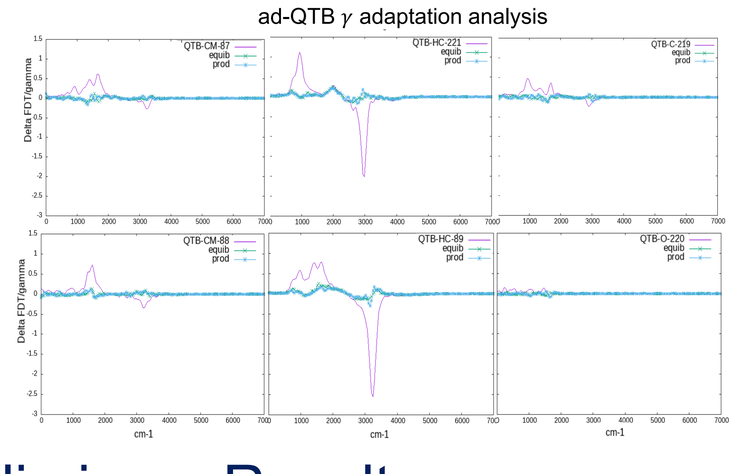
Force Field Parametrization



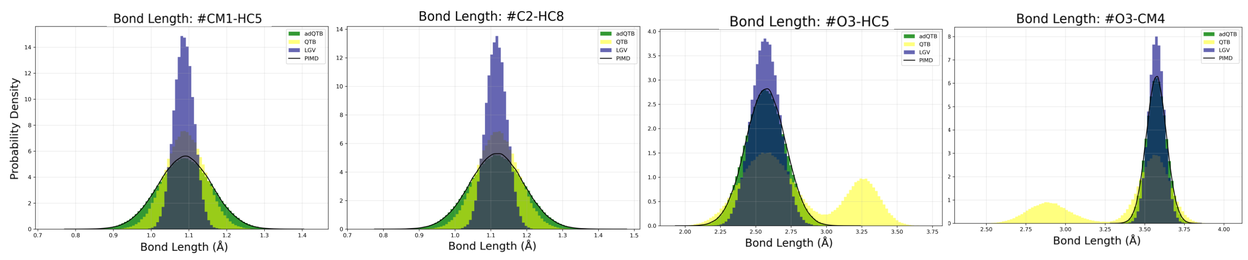


An example of leakage for a mode is shown for different values of friction coefficient (γ) in **QTB** dynamics.

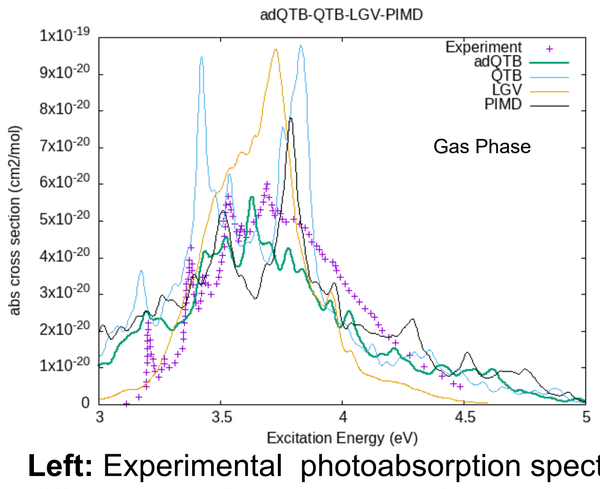
ad-QTB γ adaptation analysis

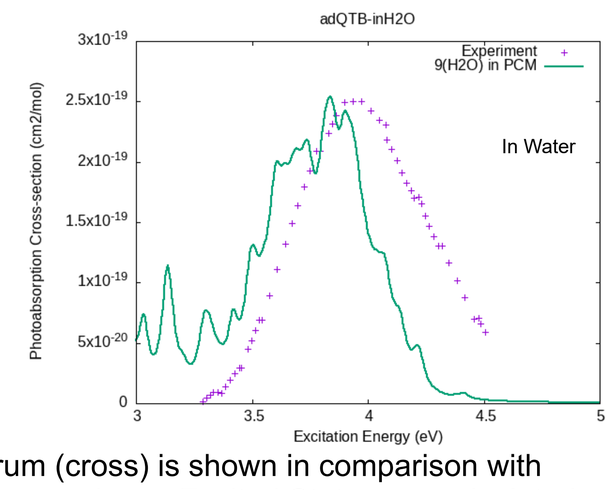


Preliminary Results



- The plot shows the broadening of mean bond lengths using Quantum MD over classical MD leading to broader spectrum, i.e. captures nuclear motion beyond classical dynamics.
- QTB appears to have artifacts due to excessive ZPE leakage.





Left: Experimental photoabsorption spectrum (cross) is shown in comparison with different sampling techniques for acrolein in gas phase. Excited States are calculated using CASPT2 with ANO-L basis.
Right: Experimental photoabsorption spectrum for acrolein in water is shown in comparison with spectrum from single ad-QTB trajectory for sampling. TDA-DFT with PBE0 is used for solvation shell of 5Å around acrolein placed in PCM.
Observations;

- ad-QTB captures most of the features in the n → π^* transitions in particular the broadening for both gas phase and acrolein in water spectra.

Conclusion and Future Works

- Benchmarked different electronic structure methods for spectroscopic properties.
- We also show effects of ZPE leakage in QTB dynamics which are compensated by using adaptive-QTB method.
- We shown that Nuclear Quantum Effects are best captured in ad-QTB method in comparison to QTB and LGV dynamics.

Further works:

- Compare gas phase and in water with PIMD.
- Perform more trajectories

Acknowledgment

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